

# Lecture 2: Study Design and Descriptive studies

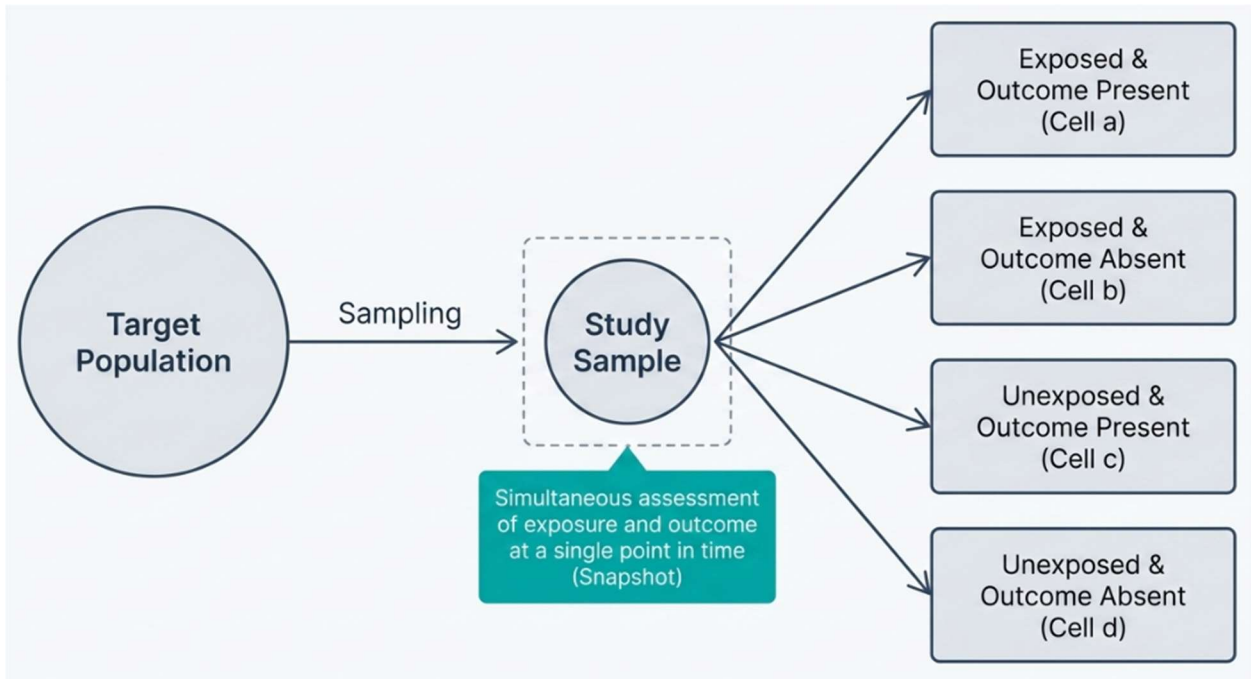
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## 1. Cross sectional surveys:

- a. collect data on variables
- b. Strengths:
  - i. Collect data not provided by other types of studies:
    - 1. Demographic data (census)
    - 2. Nutritional data
    - 3. Knowledge, attitudes and practices (KAP)
  - ii. As they are descriptive studies:
    - 1. calculate disease burden/prevalence (surveys are the best way to do that)
    - 2. Get information about variables (who is affected, where and when)
- c. Limitations:
  - i. As they are descriptive studies: cannot test hypothesis (cannot reveal association between exposure and outcome)

## 2. Cross sectional studies:

- a. Known as prevalence studies
- b. No direction, exposure and outcome are assessed simultaneously in the same point of time (like a snapshot)
- c. Calculate prevalence rates
- d. prevalence rate =
$$\frac{\text{the total number of cases old or new in a certain area}}{\text{the total number of population in the same area}} \times 100$$



Cross sectional studies flow chart

|                 |           | Outcome Status              |                            |             |
|-----------------|-----------|-----------------------------|----------------------------|-------------|
|                 |           | Outcome Present (Disease +) | Outcome Absent (Disease -) | Total       |
| Exposure Status | Exposed   | [ a ]                       | [ b ]                      | [ a + b ]   |
|                 | Unexposed | [ c ]                       | [ d ]                      | [ c + d ]   |
|                 | Total     | [ a + c ]                   | [ b + d ]                  | [ Total N ] |

Tabulated results of any study

Note: Prevalence of something among other thing is calculated as any other percentage is calculated

Example: prevalence of diabetes among obese people =

$$\frac{\text{numver of diabetc patients}}{\text{number of all obese people (whather they have diabetes or not)}} \times 100$$

e. Strengths:

i. As they are descriptive studies:

1. Generate hypothesis (show association between exposure and outcome)
2. Calculate prevalence/ case load (especially in chronic diseases)
3. easy quick and inexpensive (no follow up or comparison group)

f. Limitations:

i. Chicken egg dilemma: temporal Relation between exposure and outcome (who came first) is not established

ii. Not suitable for:

1. Rare conditions: (low possibility to be found in the chosen sample)
2. Fatal diseases: they will be dead at the time of the study (survival bias)
3. Acute and short duration diseases

iii. Cannot test hypothesis because:

1. No comparison group (as any other descriptive study)
2. Chicken egg dilemma